

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

I SARSHINI. S 192321113 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

Q1.With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.

Machine Learning Pipeline:

Raw Data → Data Collection → Data Preprocessing → Feature Extraction → Model Selection → Model Training → Model Evaluation → Deployment → Prediction.

Machine Learning System Blocks:

- Data Collection:

Collects raw data from various sources such as databases, sensors, and APIs.

- Data Preprocessing:

Cleans and prepares data by removing noise, handling missing values, and normalization.

- Feature Extraction:

Selects important features (input variables) to improve model performance.

- Model Selection:

Chooses suitable algorithms like Decision Tree, SVM, or Linear Regression.

- Model Training:

Model learns patterns from the given training data.

- Model Evaluation:

Evaluates performance using metrics like accuracy, precision, and recall.

- Deployment:

Implements the trained model into real-world applications.

- Prediction:

Model predicts output for new input data.

Supervised Learning:

- Uses labelled data.
 - Used for classification and regression.
 - Learns mapping between input and output.
- Example: Predicting Pass/Fail using student marks.

Unsupervised Learning:

- Uses unlabelled data.
 - Finds hidden patterns or clusters in data.
 - No predefined output is given.
- Example: Customer segmentation using K-Means.

Difference:

Supervised: labelled data, predicts output.

Unsupervised: unlabelled data, discovers patterns.

Q2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

- a) Describe the steps involved in training, validating, and evaluating the model.
- b) Explain how the system can assist in early warning and emergency response.

System Design:

A machine learning system is designed to predict disasters like **floods and earthquakes** using:

- Historical data (past disaster records)
- Real-time data (weather sensors, seismic data, satellite data)

Model Choice:

• Flood Prediction:

- Models: Linear Regression, Random Forest, LSTM (for time-series rainfall data)

• Earthquake Prediction:

- Models: Decision Tree, SVM, Neural Networks

• Reason:

- Time-series models (LSTM) handle continuous data
- Classification models help in risk prediction (low/high risk)

Data Preprocessing:

- Handling missing data (fill or remove values)
- Noise removal from sensor data
- Normalization and scaling
- Converting categorical data into numerical form

Q10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

System Design:

A machine learning system is designed to predict diseases like **heart disease** and **diabetes** using patient data such as:

- Age
- Blood pressure
- Glucose level
- Cholesterol
- Lifestyle factors (diet, exercise, smoking)

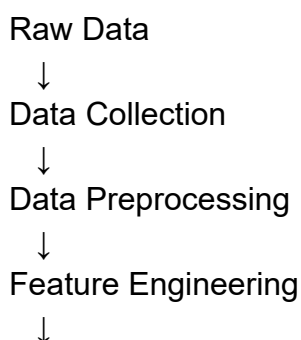
Choice of Suitable Algorithms:

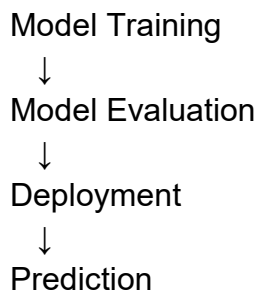
- **Logistic Regression:**
 - Used for binary classification (disease / no disease)
 - Simple and interpretable
- **Decision Tree:**
 - Easy to understand and visualize
 - Handles both numerical and categorical data
- **Random Forest:**
 - Improves accuracy by combining multiple trees
 - Reduces overfitting
- **Support Vector Machine (SVM):**
 - Effective for high-dimensional data
 - Works well for classification problems
- **Neural Networks:**
 - Captures complex relationships in medical data

Justification of Algorithm Selection:

- Logistic Regression → Best for basic disease prediction
- Decision Tree → Provides clear decision rules (useful in healthcare)
- Random Forest → High accuracy and reliability
- SVM → Good for complex patterns
- Neural Networks → Suitable for large datasets and deep analysis

Q11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.





Stages in Machine Learning Pipeline

• Data Collection

- Gathering raw data from sources like databases, sensors, APIs
- Example: customer data, medical records, images

Data Preprocessing

- Cleaning and preparing data
- Includes:
 - Handling missing values
 - Removing noise and outliers
 - Normalization and scaling

Feature Engineering

- Selecting and transforming important features
- Improves model performance
- Example: BMI from height & weight

Model Training

- Feeding training data into the algorithm
- Model learns patterns and relationships

Q15. Explain the concept of clustering in unsupervised learning.

- a) Describe the working of the K-Means algorithm with a step-by-step example.
- b) Explain how the value of K is chosen using the Elbow Method.
- c) Compare K-Means with Hierarchical Clustering.

Concept of Clustering

Clustering is an unsupervised learning technique

It groups similar data points into clusters

Data points in the same cluster are more similar to each other than to others

No labelled data is required

Example: Grouping customers based on purchasing behavior

K-Means Algorithm – Working

Steps in K-Means:

Choose K (number of clusters)

Initialize centroids randomly

Assign data points to nearest centroid

Update centroids (calculate mean of cluster points)

Repeat steps 3 & 4 until centroids do not change

Elbow Method for Choosing K

Used to find optimal number of clusters

Plot:

X-axis → Number of clusters (K)

Y-axis → Within Cluster Sum of Squares (WCSS)

Steps:

Run K-Means for different K values

Compute WCSS for each K

Plot graph

Identify “elbow point” (sharp bend)

👉 That K value is optimal

Reason:

After elbow point, improvement becomes minimal

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand